

Botond Branyicskai-Nagy

London, United Kingdom

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Education

University College London	September 2023 – (September 2024)
MSc Machine Learning	
Probabilistic and Unsupervised Learning ¹ Approximate Inference ¹ Applied ML	
Bayesian Deep Learning Supervised Learning Reinforcement Learning ²	
Applied Deep Learning ML Seminar	
Imperial College London	2020 – 2023
BSc Physics with Theoretical Physics	
Eotvos Jozsef Secondary School, Budapest	2014 – 2020
A-levels equivalent in Hungary at advanced level: Mathematics, Physics, English	

Research Experience

MSc Thesis - University College London	2024
Neural Program Synthesis for Mathematical Abstraction: modelling representations of mathematical objects as neural circuits and performing program synthesis in this space of composed functions from the learnt library of concepts. Supervised by Prof. Mirco Musolesi.	
Research Internship — Imperial College London	2023
Causal discovery using cross-validation as compared with Bayesian Model Selection (GPLVM), supervised by Mark van der Wilk and funded by Imperial UROP.	
BSc Thesis — Imperial College London	2023
Computational Evolution: Simulation and analysis of evolution by tracking agents in virtual environments. Comparison with theoretical models, exploration of optimisation algorithms.	

Other Projects

Masked Image Modelling — University College London	2024
Self-supervised learning framework: a VisionTransformer is pre-trained to reconstruct partially masked images then fine-tuned on the Oxford-IIIT Pets dataset for segmentation.	
Barabasi-Albert and Oslo Models — Imperial College London	2023
Implementation of a simulation and analysis of each model's behaviour.	
Solar Sail (Project Svarog) — Imperial College Space Society	2022
Orbital mechanics: spacecraft trajectory simulation, atmospheric effects, relativistic corrections.	
Machine learning for LHCb experiment — Imperial College London	2022
Leading ML team in project analysing decay product distribution from LHCb experiment for Standard Model comparison. Neural Networks (TensorFlow), Decision Trees.	
Laser microphone — Imperial College London	2020
Building a proof-of-concept prototype of a laser interferometer-based device capable of recovering audio from vibrations of a window from a distance.	

Awards

2nd Year Quantum Physics Article Prize – Published on Imperial Website (2022)
Young Researchers of Natural Sciences, Conference Presentation Distinction (2016)

Skills and Interests

Languages & Programming
Python - Scientific and ML libraries (TensorFlow, GPflow, JAX, PyTorch), Julia, Cluster usage
Languages
Hungarian (Native), English (Proficient - Cambridge Advanced 206), French (Intermediate)
Other Interests: Space Society, tennis, hiking, piano

¹Module part of PhD programme at the Gatsby Computational Neuroscience Unit

²Module taught by Google DeepMind

For academic references please contact:

Dr Mark van der Wilk

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Professor Daniel Waldram

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